

BioAro Drugs AI Model And Partner Services Handoff

Audience: AI, backend, product, safety, data, and frontend teams Status: Implementation handoff Canonical source: This Markdown file Last reviewed: 2026-08-18

1. Executive Summary

BioAro Drugs AI is a 360-degree everyday-health assistant combining advanced BioAro product knowledge with guidance shaped by clinical experience.

The experience has two layers:

1. Public exploration: approved product, ingredient, usage, quality, availability, and educational answers. 2. Member continuity: a persistent workspace for goals, protocol building, routine refinement, and conversation history.

The AI may interpret intent, retrieve approved content, ask structured follow-up questions, build a protocol, and identify an approved partner service. It must not diagnose, prescribe, interpret prescriptions, make drug-interaction decisions, or invent product facts.

The partner layer is a controlled commercial recommendation system. The model can identify a relevant category, but a backend policy engine decides whether a partner can be shown.

2. Current State Versus Production Target

Area	Implemented today	Production target
Public Q&A	Local approved-content retrieval	Server-side retrieval with citations
Intent interpretation	Keyword fallback; optional interpretation endpoint seam	Validated model output with goal allowlist
Protocol builder	Deterministic rules in buildProtocol	Model-assisted clarification with deterministic policy validation
Safety	Client classifier for approved retrieval	Server-side safety classifier and refusal policy
Product facts	Local editorial data and Shopify mapping	Versioned approved knowledge index
Market	Local market configuration	Server-enforced market and catalogue filters
Membership	Demo snapshot or unavailable backend	Verified account entitlement
History	Browser localStorage; resumable new records	Authenticated account storage
Images	Browser-only object URLs; no upload or interpretation	Explicit attachment service, still no prescription interpretation in V1
Partner services	Partner page exists; no AI recommendation registry	Registry plus deterministic policy endpoint

3. Product Positioning And Boundaries

Core positioning

Ask. Build. Evolve.

- Ask: understand BioAro products, ingredients, usage, availability, and approved education.
- Build: turn stated goals and routine answers into a structured starting protocol.
- Evolve: return to the workspace and refine the routine over time.

Do not position the product as

- Medical diagnosis or treatment
- A prescribing service
- A prescription or medication interpreter
- A drug interaction checker
- Biomarker or genetic analysis
- A replacement for a doctor, pharmacist, or qualified healthcare professional
- A guarantee of health outcomes

4. User Journeys

Homepage

- > Public Q&A
- > Approved answer, safety referral, or protocol handoff

Homepage goal input

- > Intent interpretation
- > Validate approved goals
- > Protocol Studio
- > Market AI workspace
- > Structured protocol
- > Optional partner policy
- > Product and partner outputs

Entry points

- Homepage free-text AI input
- Homepage goal chips and persona cards
- Homepage floating AI control
- Header AI & Protocols link
- Account Open BioAro Drugs AI action for eligible members
- Public membership page
- Direct `/:market/ai` visits
- Previous chat resume control

Homepage goal handoff

1. User enters a goal. 2. Frontend calls intent interpretation. 3. Returned goal IDs are filtered against the approved GOALS list. 4. The protocol session stores the original wording and selected goals. 5. The user continues in Protocol Studio. 6. The app routes to `/:market/ai?goals=focus,recovery` when the full workspace is selected.

5. Input Parameters

5.1 Intent request

```
interface InterpretRequest {  
  message: string;  
  selectedGoals: GoalId[];  
  market: string;  
  sessionContext?: Record<string, unknown>;  
}
```

Parameter	Importance	Validation
message	Original user wording; preserves provenance and supports interpretation	Required, trimmed, maximum 200 characters
selectedGoals	Prevents the model from discarding an explicit user choice	Array of approved goal IDs

Parameter	Importance	Validation
market	Controls product, price, availability, and partner eligibility	uk, us, ca, or ae
sessionContext	Provides non-clinical continuity	Allowlisted fields only; never raw prescription or clinical records

5.2 Public Q&A request

```
interface AskRequest {
  query: string;
  market: string;
  selectedGoals?: GoalId[];
  conversationId?: string | null;
}
```

5.3 Protocol request

```
interface ProtocolRequest {
  market: string;
  goals: GoalId[];
  answers: PartialAnswers;
  clinicalFlag: boolean;
  existingSupplements?: SupplementsAnswer;
  sessionId?: string;
}
```

5.4 Protocol answer fields

Field	Accepted values	Why it matters
energy	steady, dips, crashes	Can add daily foundational support
sleep	restful, inconsistent, poor	Can add recovery context and safety messaging; there is no approved sleep formula
training	daily, sometimes, rarely	Can add recovery or performance support
sex	female, male, prefer-not-to-say	Can route to a gender-specific multivitamin; skip remains valid
age	under-30, 30-49, 50-plus	Can add the approved D3/K2 product for 50-plus
activity	sedentary, moderate, very-active	Can affect training-related protein routing
stress	low, moderate, high	Can add approved stress/adaptogen or magnesium context
diet	omnivore, vegetarian, vegan	Selects plant protein versus whey when training is relevant
supplements	none, multivitamin, omega-3, protein	Prevents duplicate or overlapping recommendations

5.5 Clinical screener

```
interface ClinicalContext {
  clinicalFlag: boolean;
}
```

The current question asks whether the user is pregnant or breastfeeding, taking prescription medication, or managing a diagnosed condition. The answer may trigger professional-guidance copy. It must not alter the product list.

5.6 Partner context

```
interface PartnerContext {
  market: MarketCode;
  goals: GoalId[];
  nonClinicalSignals: string[];
  explicitServiceInterest?: string;
  previousPartnerIds?: string[];
}
```

Only non-clinical signals may reach partner eligibility. The backend should derive nonClinicalSignals from an allowlist rather than forwarding the complete conversation.

6. Goal Model

```
type GoalId =  
  | "energy"  
  | "longevity"  
  | "focus"  
  | "recovery"  
  | "sleep"  
  | "performance"  
  | "womens-health";
```

Rules:

- Unknown goal IDs are rejected.
- Multiple goals are preserved in selection order.
- Duplicate goals are removed.
- The model cannot introduce a product by inventing a goal.
- sleep must be handled honestly because no approved sleep formula is currently available.
- Product visibility is checked against the active market.

7. Adaptive Questions

```
interface QuestionOption {  
  value: string;  
  label: string;  
}  
  
interface ProtocolQuestion {  
  id: string;  
  field: AnswerField;  
  stage: "about" | "clinical" | "precision";  
  prompt: string;  
  options: QuestionOption[];  
  appliesTo?: GoalId[];  
}
```

A question earns a place only when at least one possible answer changes the protocol items, slot, rationale, or required disclosure. Do not ask a fixed questionnaire when the answer cannot change the result.

precision is reserved for future biomarkers, labs, genetics, or wearables and must not be presented as live capability.

8. Product Recommendation Output

```
interface ProtocolItem {  
  handle: string;  
  slot: "Morning" | "Around training" | "Evening";  
  reason: string;  
}  
  
interface Protocol {  
  items: ProtocolItem[];  
  notes: string[];  
}
```

Output importance

- handle: stable product identity; must exist in the approved catalogue.
- slot: makes the output actionable without inventing dosage.
- reason: explains the relationship to the user-stated goal or answer.
- notes: carries honest limitations and safety disclosures.

Product controls

- The model never owns price, inventory, or availability.

- Handles are validated against Shopify or approved catalogue data.
- Product details are retrieved by handle after model output validation.
- Reasons must be grounded in approved product copy.
- Doses, ingredients, evidence, and claims must come from approved sources.
- Existing supplement overlap can remove a product.
- No unrelated product may be substituted for an unavailable goal.

9. Public Q&A API

Current frontend boundary:

```
type AskResult =
  | { kind: "answer"; answers: AskAnswer[] }
  | { kind: "needs-quiz" }
  | { kind: "sensitive" }
  | { kind: "no-match" };

interface AskAnswer {
  question: string;
  answer: string;
  source: {
    kind: "product" | "faq" | "journal";
    label: string;
    href: string;
  };
}
```

Production endpoint:

```
POST /api/bioaro-ai/ask
```

Example request:

```
{
  "query": "How much NMN is in LONGevity+?",
  "market": "US",
  "selectedGoals": ["longevity"],
  "conversationId": null
}
```

Example response:

```
{
  "requestId": "req_123",
  "kind": "answer",
  "answers": [{
    "question": "How much NMN is in LONGevity+?",
    "answer": "Approved source text...",
    "source": {
      "kind": "product",
      "label": "LONGevity+",
      "href": "/us/products/longevity-plus"
    }
  }],
  "partnerRecommendations": [],
  "safety": {
    "status": "allowed",
    "clinicalReviewRequired": false,
    "disclaimer": "BioAro Drugs AI provides general product guidance..."
  },
  "provenance": {
    "model": "approved-retrieval-v1",
    "retrievalSources": ["product:longevity-plus:ingredient:nmn"],
    "policyVersion": "ai-policy-v1"
  }
}
```

Response states:

- answer: approved content was found.
- needs-quiz: user asked for a personal product choice.
- sensitive: professional guidance is required.
- no-match: no grounded answer was found.
- error: service failure; client may offer retry.

10. Intent Interpretation API

POST /api/bioaro-ai/interpret

```
interface InterpretResponse {
  interpretedGoals: GoalId[];
  confidence: number;
  summary: string;
  suggestedNextQuestion?: string;
  builderContext?: { primaryGoals: GoalId[] };
}
```

Validation requirements:

- Confidence must be numeric and bounded from 0 to 1.
- interpretedGoals must contain only approved IDs.
- summary must be a plain acknowledgement, not a health claim.
- Raw model text must not be rendered without schema validation.
- Prompt injection must not override system policy or catalogue boundaries.

11. Combined AI Response

```
interface BioAroAiResponse {
  requestId: string;
  kind: "answer" | "protocol" | "clarification" | "sensitive" | "no-match" | "error";
  summary?: string;
  answers?: AskAnswer[];
  protocol?: Protocol;
  nextQuestion?: ProtocolQuestion;
  partnerRecommendations?: PartnerRecommendation[];
  safety: {
    status: "allowed" | "referral" | "blocked";
    clinicalReviewRequired: boolean;
    disclaimer: string;
  };
  provenance: {
    model: string;
    retrievalSources: string[];
    policyVersion: string;
  };
}
```

Output	Importance
requestId	Trace support incidents without exposing user content
kind	Selects the correct UI state
summary	Safe user-facing acknowledgement
answers	Grounded, cited information
protocol	Structured routine output
nextQuestion	Continuation of the builder
partnerRecommendations	Optional controlled ecosystem promotion
safety	Mandatory safety and disclosure boundary
provenance	Auditability and source transparency

12. Partner-Service Promotion

Registry

```
interface PartnerService {
  id: string;
  name: string;
  description: string;
  category:
    | "sports-performance"
    | "healthy-ageing"
```

```

    | "testing-and-insights"
    | "professional-support"
    | "other";
  approvedMarkets: MarketCode[];
  approvedGoals: GoalId[];
  approvedAnswerSignals: string[];
  excludedSignals: string[];
  url: string;
  ctaLabel: string;
  disclosure: string;
  status: "active" | "draft" | "paused";
}

```

Initial registry records:

- biosports: draft until approved service description, URL, markets, and rules exist.
- biogeivity: draft until approved service description, URL, markets, and rules exist.
- bioaro-labs: draft until approved service description, URL, markets, and rules exist.

Matching policy

Allowed signals:

- Explicit goals
- Training and performance interest
- Healthy-ageing interest
- Interest in evidence or testing
- Explicit service requests
- Approved non-clinical lifestyle answers

Blocked signals:

- Diagnoses or symptoms
- Medication or prescription content
- Pregnancy or breastfeeding
- Clinical flags
- Biomarkers or genetics
- Inferred medical risk
- Protected health information

The final eligibility decision must be made by backend policy, not by model confidence alone.

Partner output

```

interface PartnerRecommendation {
  partnerId: string;
  name: string;
  reason: string;
  ctaLabel: string;
  href: string;
  disclosure: string;
  confidence: number;
  eligibility: {
    market: boolean;
    goal: boolean;
    policy: boolean;
  };
}

```

Rules:

- Maximum one or two partner recommendations per response.
- Always label partner content clearly.
- Show partner cards below or beside BioAro product answers, never as the primary medical-looking answer.
- Suppress on sensitive requests, clinical flags, unsupported markets, missing metadata, or missing disclosure.
- Do not imply clinical endorsement, guaranteed results, or required purchase.

Partner endpoint

POST /api/bioaro-ai/partner-recommendations

Request:

```
{
  "market": "US",
  "goals": ["performance", "recovery"],
  "answers": {
    "training": "daily",
    "activity": "very-active"
  },
  "explicitInterest": null,
  "safetyStatus": "allowed"
}
```

Response:

```
{
  "recommendations": [],
  "suppressed": [],
  "policyVersion": "partner-policy-v1"
}
```

Every displayed recommendation must include:

Partner service suggested based on your stated interests. BioAro Drugs may have a commercial relationship with this partner.

13. Safety Pipeline

Input

- > Auth and market validation
- > Prompt injection check
- > Sensitive and clinical classifier
- > Approved retrieval
- > Protocol engine
- > Partner eligibility policy
- > Schema, citation, and disclosure validation
- > Client response

Production safety must be server-side. The current client denylist is acceptable only for retrieving already-approved copy; it is not sufficient for generated health text.

Partner recommendations must be disabled if safety status is referral, clinicalFlag is true, prescription or medication context exists, partner metadata is missing, market is unsupported, or disclosure is missing.

14. Retrieval And Knowledge Model

Approved sources:

- Shopify product data and metafields
- Product ingredients, amounts, supplement facts, warnings, and quality claims
- FAQs and approved journal content
- Market availability and prices
- Partner registry and partner-approved service descriptions

Recommended retrieval document:

```
{
  "documentId": "product:longevity-plus:ingredient:nmn",
  "sourceType": "product_ingredient",
  "productHandle": "longevity-plus",
  "market": ["GB", "US", "CA", "AE"],
  "content": "Approved source text...",
  "sourceUrl": "/us/products/longevity-plus",
  "sourceVersion": "shopify_updated_at",
  "allowedUses": ["product_question", "protocol_reason"],
  "blockedUses": ["diagnosis", "treatment", "drug_interaction"],
  "visibility": "published"
}
```

Partner documents must be indexed separately from product documents and must include partnerId, commercial status, market restrictions, and disclosure text.

15. Image And Attachment Handling

Current behavior

- User selects label or prescription before choosing a file.
- The file becomes a browser object URL.
- No upload, OCR, or model interpretation occurs.
- Images are excluded from local history.
- The object URL is revoked when the thread is cleared.

Proposed future boundary

POST /api/bioaro-ai/attachments

```
interface AttachmentRequest {
  intent: "label" | "prescription";
  market: MarketCode;
  conversationId: string;
  contentType: string;
  size: number;
  checksum: string;
}

interface AttachmentResponse {
  attachmentId: string;
  intent: "label" | "prescription";
  status: "stored" | "rejected" | "expired";
  expiresAt: string;
  interpretationStatus: "not_requested";
}
```

Prescription attachments remain storage/reference-only until a separate approved clinical workflow exists.

16. Advanced AI And Membership APIs

Eligibility predicate:

```
membership.status == "active"
AND membership.tierId in ["essential", "plus"]
```

Proposed endpoints:

```
GET /api/member/snapshot
POST /api/member/conversations
GET /api/member/conversations
GET /api/member/conversations/:id
POST /api/member/conversations/:id/messages
POST /api/member/protocols
GET /api/member/protocols
POST /api/member/attachments
```

Every endpoint requires authenticated account ownership and market isolation. A successful checkout must not grant AI access until the backend membership snapshot confirms eligibility.

17. Conversation History

Current local shape:

```
interface Conversation {
  id: string;
  market: string;
  title: string;
  startedAt: string;
  updatedAt: string;
  turns: Turn[];
  session?: ProtocolSession;
}
```

Current turn types:

- ai: assistant text
- you: user text

- ask: structured Q&A question plus result
- image: local attachment reference, stripped before persistence

New records can be resumed with goals and answers. Older records without saved session state restore text safely. Production history must move to authenticated account storage and must not place health or attachment content in URLs, analytics, or unencrypted client storage.

18. Shopify And Membership Commerce

Membership page

- > Shopify variant and selling plan
- > First-party checkout
- > Shopify purchase
- > Membership success page
- > Backend membership refresh
- > Verified entitlement
- > Advanced AI access

Shopify owns price, currency, selling plan, billing interval, availability, and market eligibility. The AI never controls commerce or entitlement.

19. Error And Fallback Matrix

Failure	User-visible result	System behavior
Empty input	Ask for a question or goal	Do not call the model
Input too long	Shorten the message	Reject before request
Unknown goal	Clarification	Do not create a product
No retrieval match	No approved answer found	Offer search, quiz, or support
Sensitive request	Professional-guidance response	Do not retrieve product advice or partners
Invalid model JSON	Temporary error	Log schema failure and use deterministic fallback
Unknown product handle	No product shown	Reject output and log grounding failure
Unsupported market	Market unavailable	Do not show price or partner CTA
Partner metadata missing	No partner card	Suppress recommendation
Missing disclosure	No partner card	Reject partner output
Membership unavailable	Retry/support gate	Do not unlock workspace
Attachment rejected	Explain file limitation	Do not persist or interpret
Timeout or rate limit	Retry later	Apply backoff and rate limiting
Conversation save failure	Continue current session	Make persistence failure visible to member

20. Observability And Evaluation

Track non-sensitive metrics:

- Intent interpretation success and rejection rate
- Retrieval answer and no-match rate
- Safety referral rate
- Protocol completion and abandonment rate
- Partner recommendation and suppression rate
- Partner CTA clicks and conversion
- Invalid structured-output rate
- Product grounding failures

- Conversation resume success
- Latency and errors by endpoint and market

Do not send raw health answers, prescriptions, medication details, or clinical flags to analytics by default.

Evaluation sets must include:

- Single and multi-goal prompts
- Ambiguous goals
- Product and ingredient-dose questions
- Availability questions
- Recommendation requests
- Sensitive and emergency prompts
- Prompt-injection attempts
- Existing supplement overlap
- Sleep-goal limitations
- Partner matching and suppression cases
- Unsupported markets
- Invalid model output
- Cross-account and cross-market isolation

21. Visual References

The handoff should be reviewed alongside these repository visuals:

These images are visual references only. They are not model outputs and must not be treated as clinical evidence. The implementation package should also include current screenshots of /us/ai, the Previous chats modal, and /us/membership when the final backend build is reviewed.

22. Implementation Checklist

1. Confirm backend hosting, authentication, and account identity binding. 2. Define the approved partner registry. 3. Supply approved BioSports, Biogevity, and BioAro Labs descriptions, URLs, markets, and disclosures. 4. Implement server-side safety and prompt-injection classification. 5. Ingest Shopify and approved editorial content into a versioned retrieval index. 6. Implement /api/bioaro-ai/interpret. 7. Implement /api/bioaro-ai/ask. 8. Implement protocol recommendation and question selection. 9. Implement partner eligibility policy and /api/bioaro-ai/partner-recommendations. 10. Add structured-output, product-handle, source, and disclosure validation. 11. Add membership entitlement checks. 12. Add authenticated conversation and protocol persistence. 13. Add attachment storage boundary without prescription interpretation. 14. Add rate limiting, audit logging, and non-sensitive observability. 15. Connect frontend feature flags and fallback behavior. 16. Run safety, grounding, partner, privacy, and regression evaluations. 17. Launch behind a controlled feature flag.

23. Interface Inventory

Existing frontend contracts

- InterpretRequest
- InterpretResponse
- InterpretResult
- AskResult
- AskAnswer
- ProtocolQuestion
- ProtocolAnswers
- ProtocolItem

- Protocol
- Conversation
- Turn
- MemberRequest
- MemberSnapshot
- MembershipOffer

Proposed backend contracts

- AskRequest
- ProtocolRequest
- PartnerService
- PartnerContext
- PartnerRecommendation
- BioAroAiResponse
- AttachmentRequest
- AttachmentResponse

Sensitive/private data

- Clinical flags
- Prescription references
- Medication context
- Account identity
- Member conversations
- Attachment metadata

These fields must never be used for uncontrolled partner promotion or exposed to analytics by default.

24. Final Safety Position

BioAro Drugs AI can help users understand approved information, structure goals, assemble a routine, return to previous work, and discover approved BioAro partner services.

It cannot determine what is medically right for a person. Partner promotion is a transparent service-discovery layer governed by approved metadata, market policy, safety state, and commercial disclosure.

The production launch is complete only when every displayed answer, product, price, source, partner, and entitlement is traceable to an approved system of record.